

Dinesh Kumar Yadav

Data Science | Backend Engineering | Web Development | Deployment

Varanasi, Uttar Pradesh, India

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in **Data Science, Machine Learning, Backend Engineering, and Web Development**. Skilled in building end-to-end solutions across **AI/ML, backend APIs, web applications, databases, and production deployment** using Python, FastAPI, React/Next.js, SQL, and Generative AI. Strong problem-solving mindset with an interest in **Forward Deployed Engineering**, applying data, software engineering, and AI to solve real-world problems.

TECHNICAL SKILLS

Programming: Python, SQL, C, C++, Java, JavaScript, TypeScript

Data Science & ML: NumPy, Pandas, Scikit-learn, EDA, Data Preprocessing, Feature Engineering, Regression, Classification, Clustering, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning & Computer Vision: TensorFlow, Keras, PyTorch, CNN, U-Net, Transfer Learning, Image Segmentation, OpenCV, Data Augmentation

Generative AI: LLMs, RAG, Prompt Engineering, Vector Embeddings, Conversational AI, AI Chatbots, LangChain

Frontend: React.js, Next.js, JavaScript, TypeScript, HTML, CSS, Tailwind CSS

Backend & APIs: Python, FastAPI, REST APIs, JWT Authentication, API Integration, Backend Architecture

Databases: MySQL, PostgreSQL, MongoDB, SQL, Database Design, Query Optimization

Deployment & Developer Tools: Git, GitHub, Vercel, Docker, Postman, Google Colab, Jupyter Notebook, VS Code

Quantum ML: Quantum Machine Learning Fundamentals, Variational Quantum Circuits, Qiskit, PennyLane

RESEARCH & INTERNSHIP EXPERIENCE

Research Intern – Quantum Machine Learning

August 2026 – Present

Symbiosis International University (SIU)

Pune, India (Remote)

- Conducting applied research on **Quantum Machine Learning**, including variational quantum circuits and hybrid quantum-classical models for classification tasks.
- Implementing and benchmarking quantum ML algorithms in Python against classical machine learning baselines.
- Working with quantum computing frameworks to encode classical datasets into quantum states and analyze model behaviour and computational trade-offs.

Research Intern – Medical Image Segmentation

25 June – 25 July 2026

Symbiosis International University (SIU)

Pune, India (Remote)

- Developed a **U-Net based deep learning pipeline** for automated **Meibomian gland segmentation** from infrared meibography images.
- Performed medical image preprocessing including normalization, contrast enhancement, resizing, data augmentation, and dataset preparation.
- Trained and evaluated segmentation architectures using **Dice Coefficient, IoU, Precision, Recall, F1-score, and pixel-level evaluation**.
- Analyzed model predictions and segmentation failure cases caused by image noise and ambiguous gland boundaries.

Freelance Developer – AI, Backend & Web Applications

Dec 2024 – Present

Independent

India

- Developing end-to-end applications combining **AI/ML models, backend APIs, databases, web interfaces, and production deployment**.
- Building REST APIs using **Python and FastAPI** for AI-powered applications and frontend-backend integration.
- Integrating Generative AI and chatbot capabilities into web applications for conversational interaction and automated information access.
- Working across the complete application lifecycle, including solution design, development, API integration, testing, deployment, and maintenance.

PROJECTS

MCAET College Website & AI Chatbot | *Next.js, React, AI, Chatbot, Vercel* | [Live Website](#) 2024

- Developed and deployed the **official web platform for Mahamaya College of Agricultural Engineering & Technology (MCAET)**, currently used by the college for institutional information and student-facing access.
- Integrated an **AI-powered chatbot** to provide conversational access to college information and frequently requested institutional information.
- Implemented responsive web interfaces and integrated application components required for real-world institutional use.
- Managed the application from development and integration through **production deployment and live usage**.

GlowMart India – AI Product Recommendation Platform | *Python, ML, FastAPI, Web Development* 2024

- Developed an AI-enabled cosmetic e-commerce platform with an ML-based product recommendation system.
- Built backend functionality for authentication, product management, database operations, and ML recommendation integration.

Facial Emotion Recognition | *CNN, TensorFlow/Keras, OpenCV* January 2026

- Developed a CNN-based computer vision model to classify six facial emotion categories using 4,500+ grayscale images.
- Applied image preprocessing, data augmentation, and regularization techniques to improve model generalization.
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Cattle Breed Classification | *CNN, Transfer Learning, Computer Vision* July 2025

- Developed a deep learning image classification pipeline for identifying multiple cattle breeds from image data.
- Applied preprocessing, augmentation, CNN architectures, and transfer learning techniques for model training and evaluation.

MCAET Examination Portal & AI Assistant | *Web Development, AI, API Integration, Vercel* | [Live Portal](#) 2026

- Developed and deployed the **MCAET Examination Portal**, currently used by the college for examination-related information and student services.
- Integrated an **AI chatbot assistant** to enable natural-language interaction with examination-related information.
- Implemented application logic and integrations for communication between the user interface, data sources, and AI-powered functionality.
- Handled development, testing, integration, and **production deployment** of the web application.

EDUCATION

Acharya Narendra Deva University of Agriculture & Technology (ANDUAT) Ayodhya, Uttar Pradesh
Bachelor of Technology (B.Tech) – Computer Science and Engineering Aug 2024 – May 2028

- Currently in 3rd Year (V Semester) | Expected Graduation: May 2028
- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Probability & Statistics, Linear Algebra, Operating Systems, Object-Oriented Programming, Machine Learning.

CERTIFICATIONS & ACHIEVEMENTS

- **2× Smart India Hackathon (SIH) Internal Hackathon Winner** – College Level
- Python for Data Science – Simplilearn
- AI Predict (Python) Workshop – SkillManthan Technologies Pvt. Ltd.